

Basics of Electrical Engineering

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1 DC Circuits

1.1 Voltage:

The integral of electric field over a length of wire gives the voltage (or potential difference) between the ends of the wire length.

Voltage is measured in Volts (V)

$$V = \int \vec{E} \cdot d\vec{l}$$

1.2 Power:

Power is the rate of which the electrical energy is used, or produced.

Power is measured in Watts (W)

$$P = \frac{dW}{dt} = \frac{dW}{dq} \cdot \frac{dq}{dt} = V \cdot i$$

1.3 Sign Convention:

1. **Active Sign Convention:** When the current **goes out** through the positive terminal and $P = -Vi$ **OR** when it **enters** the positive terminal and $P = +Vi$
2. **Passive Sign Convention:** When the current enters the positive terminal and $P = -Vi$ **OR** when it **goes out** through the positive terminal and $P = +Vi$

1.4 Ohm's Law:

1. **Extinsic Form:** $i = Vg \implies V = iR$
2. **Intrinsic Form:** $J = \sigma E \implies E = \rho J$
3. **Characteristic Graph:** It is the current vs voltage graph, describing the power. The characteristic graph for a resistor is a straight line passing from the origin.

4. **Voltage Source:** It gives a predetermined voltage irrespective of whatever it is connected to $[V = \text{constant}]$. A closed switch behaves like a voltage source. $[V = 0]$.
5. **Current Source:** It gives a predetermined current irrespective of whatever it is connected to $[i = \text{constant}]$. An open switch behaves like a current source. $[i = 0]$.
6. You cannot connect two current sources in series, and two voltage sources in parallel.

1.5 Kirchhoff's Laws:

1.5.1 Kirchhoff's Current Law (KCL):

It is the Law of Conservation of Charge. It states that current at a node is zero. Current Entering a junction = Current Exiting from that junction ($\sum i = 0$)

1.5.2 Kirchhoff's Voltage Law (KVL):

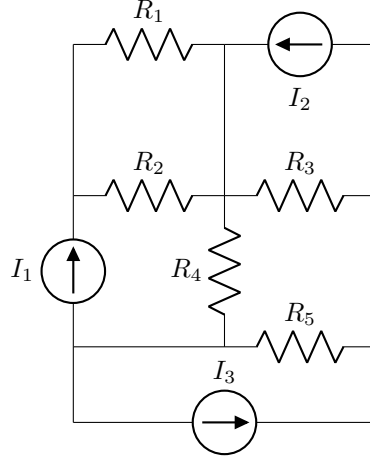
It is the Law of Conservation of Energy. It states that voltage in a loop is zero. ($\sum V + \sum iR = 0$)

1.6 Series and Parallel Resistances:

1. For two resistances R_1 and R_2 , with voltages V_1 and V_2 and common current i , in series, $\frac{V_1}{R_1} = \frac{V_2}{R_2} \implies V_1 = \frac{VR_1}{R_1+R_2}; V_2 = \frac{VR_2}{R_1+R_2}$
2. For two conductances G_1 and G_2 , with currents i_1 and i_2 and common voltage V , in parallel, $\frac{i_1}{G_1} = \frac{i_2}{G_2} \implies i_1 = \frac{iG_1}{G_1+G_2}; i_2 = \frac{iG_2}{G_1+G_2}$

1.7 Nodal Analysis

1. In this method of solving a circuit, we find the **voltages at each node** (or junction) by using **KCL**
2. We write the currents through current sources on the left side of the equations and currents through components on the right side.
3. For $k+1$ nodes, k linear algebraic equations are to be written.
4. Example Circuit:



At V_1 : $I_1 = (V_1 - V_2) \cdot (g_1 + g_2)$

At V_2 : $I_2 = (V_1 - V_2) \cdot (g_1 + g_2) + (V_2 - V_3) \cdot g_3 + V_2 \cdot g_4$

At V_3 : $I_3 - I_2 = (V_3 - V_2) \cdot g_3 + V_3 \cdot g_5$

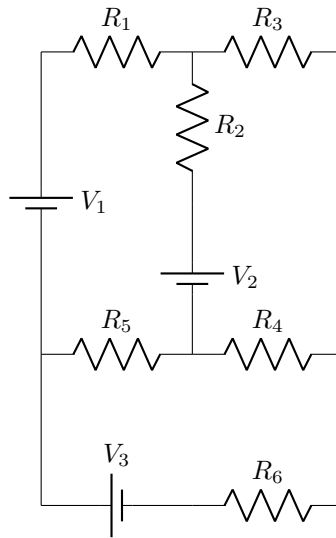
5. We can write these equations by simply observing the circuit in **matrix form** of $\mathbf{I} = \mathbf{GV}$ where $\mathbf{I}$ is the current matrix, $\mathbf{G}$ is the conductance matrix, and $\mathbf{V}$ is the voltage matrix.
6. $\mathbf{I}$ and $\mathbf{V}$ are $n \times 1$ column matrices and $\mathbf{G}$ is a $n \times n$ square matrix, where n is the number of nodes in the circuit.
7. For $\mathbf{G}$ matrix, the g_{ii}^{th} place is the sum of all conductances connected to the i^{th} node. The g_{ij}^{th} place is the negative of sum of all conductances common to i^{th} and j^{th} nodes.
8. The $\mathbf{G}$ matrix is symmetric, meaning $g_{ij} = g_{ji}$
9. The matrix equation $\mathbf{I} = \mathbf{GV}$ for the above example problem is:

$$\begin{bmatrix} I_1 \\ I_2 \\ I_3 - I_2 \end{bmatrix} = \begin{bmatrix} g_1 + g_2 & -(g_1 + g_2) & 0 \\ -(g_1 + g_2) & g_1 + g_2 + g_3 + g_4 & -g_3 \\ 0 & -g_3 & g_3 + g_5 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix}$$

10. If a voltage source is found in the circuit, the end nodes of the voltage source is considered to be only one node by applying KVL across them, calling it a "supernode".

1.8 Mesh Analysis

1. In this method of solving a circuit, we find the **currents at each mesh** (or loop) by using **KVL**.
2. We write the KVL equation ($\sum V + \sum iR = 0$) for each mesh and decide currents $i_1, i_2, \dots, i_n$ for each mesh.
3. For k meshes, k linear algebraic equations are to be written.
4. Example Circuit:



For mesh 1, $-V_1 + R_1 i_1 + R_2 \cdot (i_1 - i_2) + V_3 + R_5 \cdot (i_1 - i_3) = 0$

For mesh 2, $-V_2 + R_2 \cdot (i_2 - i_1) + R_3 \cdot i_2 + R_4 \cdot (i_2 - i_3) = 0$

For mesh 3, $R_5 \cdot (i_3 - i_1) + R_4 \cdot (i_3 - i_2) + R_6 \cdot i_3 - V_3 = 0$

5. We can write these equations by simply observing the circuit in **matrix form** of $\mathbf{V}=\mathbf{R}\mathbf{I}$ where V is the voltage matrix, R is the resistance matrix, and I is the current matrix.
6. V and I are nx1 column matrices and R is a nxn square matrix, where n is the number of meshes in the circuit.
7. For R matrix, the r_{ii}^{th} place is the sum of all resistances connected to the i^{th} mesh. The r_{ij}^{th} place is the negative of sum of all resistances common to i^{th} and j^{th} meshes.
8. The R matrix is symmetric, meaning $r_{ij} = r_{ji}$

9. The matrix equation $\mathbf{V}=\mathbf{RI}$ for the above example problem is:

$$\begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} R_1 + R_2 + R_5 & -R_2 & -R_5 \\ -R_2 & R_2 + R_3 + R_4 & -R_4 \\ -R_5 & -R_4 & R_4 + R_5 + R_6 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix}$$

10. If current source is found in the circuit, the connecting meshes of the current source is considered to be only one mesh by applying KCL along them, calling it a "supermesh".

1.9 Linear Networks

Assume a network having forcing functions F_1 and F_2 , and responses V_1 and V_2 . Then the network is said to be **linear** if:

1. For $(F_1, F_2) = (0, 0) \implies V = 0$
2. For $(F_1, F_2) = (F_1, 0) \implies V = V_1$
3. For $(F_1, F_2) = (k \cdot F_1, 0) \implies V = k \cdot V_1$
4. For $(F_1, F_2) = (0, F_2) \implies V = V_2$
5. For $(F_1, F_2) = (0, k' \cdot F_2) \implies V = k' \cdot V_2$
6. For $(F_1, F_2) = (k \cdot F_1, k' \cdot F_2) \implies V = k \cdot V_1 + k' \cdot V_2$

To solve a linear network, we have a few theorems:

1.9.1 Superposition Theorem:

To solve a linear network using superposition theorem:

1. Turn off all the independent sources except one.
2. Measure the current/voltage from that source.
3. Repeat for all independent sources in the network.
4. The total current/voltage is the sum of all the individual current/voltages.

1.9.2 Source Transformation:

A source transformation is the process of replacing a voltage source V in series with a resistor R by a current source i in parallel with a resistor R , or vice versa.

1.9.3 Thevenin and Norton Equivalent:

Here, we find two parameters: Thevenin Voltage (V_{th}) and Thevenin Resistance (R_{th}), from which we could find the Norton Current (I_N).

The steps to find the Thevenin equivalents is:

1. Convert all current sources into voltage sources.
2. Remove all the voltage sources from the circuit and solve for the equivalent resistance across the required nodes. This is the Thevenin resistance.
3. The final voltage across those two nodes is the Thevenin voltage.
4. The Norton current is the Thevenin voltage divided by the Thevenin resistance.

1.9.4 Maximum Power Transmission Theorem:

Maximum power is dissipated across a load resistor if and only if it is equal to the equivalent internal resistance.

1.10 Transient Analysis

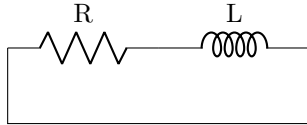
1. Capacitor:

$$i = C \frac{dV}{dt} \Rightarrow V(t) = V_0 + \frac{1}{C} \int_0^t i dt$$

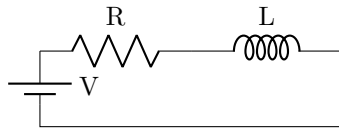
2. Capacitor:

$$V = L \frac{di}{dt} \Rightarrow i(t) = i_0 + \frac{1}{L} \int_0^t V dt$$

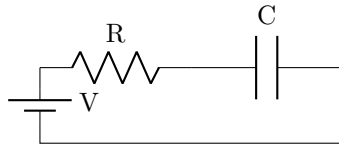
3. Equations of Circuits:



$$i_L R + L \frac{di_L}{dt} = 0 \Rightarrow \frac{di_L}{dt} + i_L \frac{R}{L} = 0$$



$$i_L R + L \frac{di_L}{dt} = V \Rightarrow \frac{di_L}{dt} + i_L \frac{R}{L} = \frac{V}{L}$$



$$V = R \frac{dq}{dt} + \frac{q}{C} \Rightarrow \frac{dq}{dt} + q \frac{1}{RC} = \frac{V}{R}$$

All these equations are of type $\frac{dy}{dx} + Py = Q$

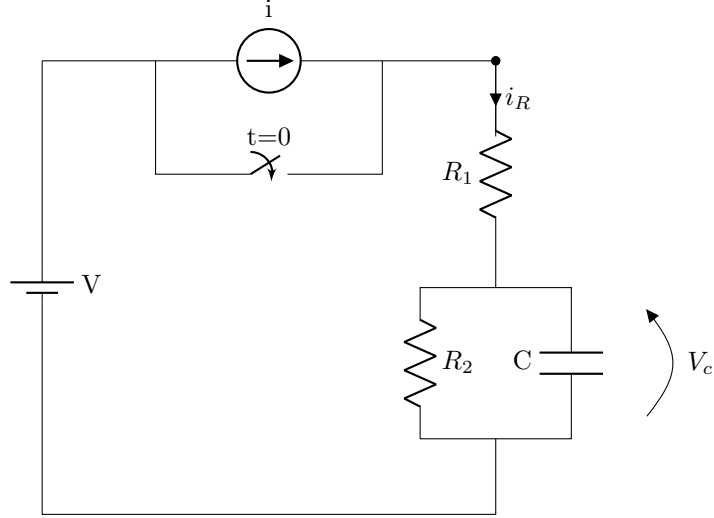
Solutions to these equations is: $y(t) = A + Be^{-Pt} = A + Be^{t/\tau}$

Initial State: $y(0) = A+B$, and Steady State: $y(\infty) = A$

Hence, $y(t) = y(\infty) + [y(0) - y(\infty)]e^{-t/\tau}$

1.10.1 First Order Circuits:

1. In such circuits, only one independent energy storage element is present.
2. **Question:** Find $i_R(t), t \geq 0$ and $V_c(t), t \geq 0$



At steady state, C act like a open circuit.

$$V_c(0^-) = V_{R_2} = iR_2 \implies V_c(0^+) = iR_2$$

Now,

$$V_c(\infty) = V \left(\frac{R_2}{R_1 + R_2} \right)$$

Also,

$$\tau = R_{eq}C = \left(\frac{1}{R_1} + \frac{1}{R_2} \right)^{-1} C = \frac{R_1 R_2 C}{R_1 + R_2}$$

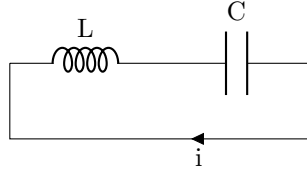
$\therefore V_c(t) = V_c(\infty) + [V_c(0) - V_c(\infty)]e^{-t/\tau}$ Hence,

$$V_c(t) = \frac{V R_2}{R_1 + R_2} + \left(iR_2 - V_c(t) = \frac{V R_2}{R_1 + R_2} \right) e^{\frac{-(R_1 + R_2)t}{R_1 R_2 C}}$$

Now, $i_R(0^+) = \frac{1}{R_1}(V - V_c(0))$ and $i(\infty) = \frac{1}{R_1 + R_2}V$ Hence,

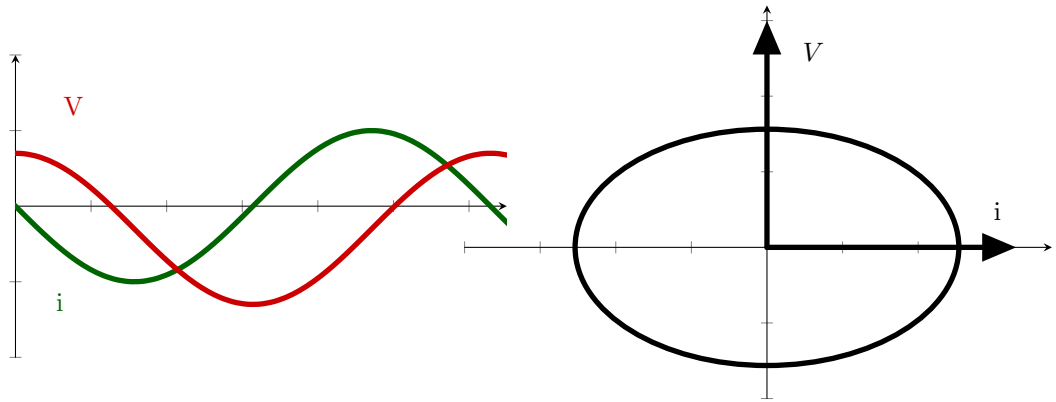
$$i_R(t) = \frac{V}{R_1 + R_2} + \left(\frac{V - V_c(0)}{R_1} - \frac{V}{R_1 + R_2} \right) e^{\frac{-(R_1 + R_2)t}{R_1 R_2 C}}$$

1.10.2 Second Order Circuits:



$$L \frac{di}{dt} + V = 0 \implies L \frac{d}{dt} \left(C \frac{dV}{dt} \right) + V = 0$$

$$LC \frac{d^2 V}{dt^2} + V = 0 \implies \boxed{\frac{d^2 V}{dt^2} + \frac{1}{LC} V = 0}$$



Underdamped System:

$$y = e^{-\alpha t} [A_1 \cos(\omega_d t) + A_2 \sin(\omega_d t)]$$

Here we have $\alpha = \frac{1}{2L}R$ and $\omega_d = \sqrt{\omega_0^2 - \alpha^2}$

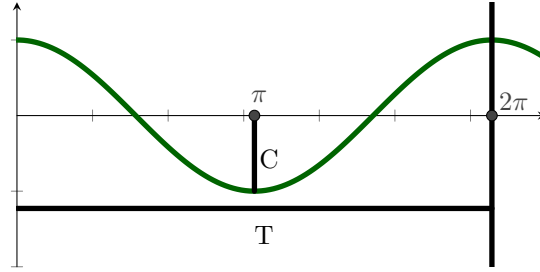
Overdamped System:

$$y = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

2 AC Circuits

2.1 Introduction:

Consider a sinusoidally varying wave $V = C \cos(\omega t - \theta)$



Now, $V = C \cos(\omega t - \theta)$

$$V = C \cos \theta \cos(\omega t) - C \sin \theta \sin(\omega t)$$

$$\therefore V = A \cos(\omega t) - B \sin(\omega t)$$

where $A = C \cos \theta$ and $B = C \sin \theta$

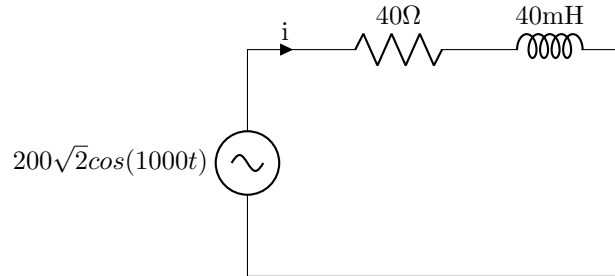
$$\therefore C = \sqrt{A^2 + B^2} \text{ and } \theta = \tan^{-1}(-B/A)$$

Similarly,

$$V = \operatorname{Re}(C e^{j\omega t} e^{j\theta})$$

2.2 Linear Elements in AC Circuits:

1. **Resistors:** $V = IR \implies \bar{V} = \bar{I}R$
2. **Inductors:** $V = L \frac{dI}{dt} \implies \bar{V} = L(j\omega \bar{I}) = j\omega L \bar{I}$
3. **Capacitors:** $I = C \frac{dV}{dt} \implies \bar{I} = C(j\omega \bar{V}) \implies \bar{V} = \frac{1}{j\omega C} \bar{I}$
4. **Question:** Find current in the circuit:

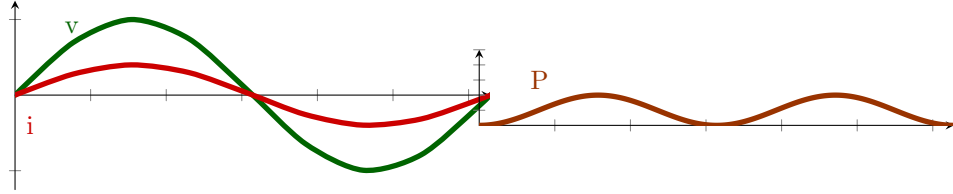


We have, $\omega = 1000 \text{ rad/s}$, $\bar{V} = 200 \angle 0^\circ$, $R = 40 \Omega$, $\chi_L = 40 \Omega$
 $\implies z = R + j\chi_L = 40 + 40j = 40\sqrt{2} \angle 45^\circ$

$$\implies \bar{I} = \frac{\bar{V}}{z} = \frac{200 \angle 0^\circ}{40\sqrt{2} \angle 45^\circ} = \frac{5}{\sqrt{2}} \angle -45^\circ \text{ A}$$

2.3 Power:

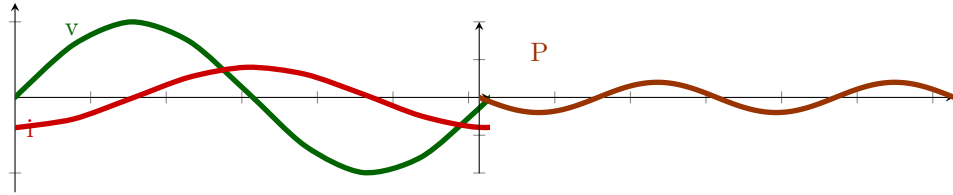
1. For Resistors:



Instantaneous Power, $p = vi$

Average Power, $P = VI$

2. For Inductors:



Average (Real) Power, $P = 0$. Inductor stores energy cyclically.

Reactive Power, $Q = VI$

3. Let $v = \sqrt{2}V \sin(\omega t)$, $i = \sqrt{2}I \sin(\omega t - \theta)$

Now,

$$\begin{aligned} p &= 2VI[\sin(\omega t)\sin(\omega t - \theta)] \\ p &= 2VI[\sin(\omega t)(\sin(\omega t)\cos\theta - \cos(\omega t)\sin\theta)] \\ p &= 2VI[(\sin^2(\omega t)\cos\theta - \cos(\omega t)\sin(\omega t)\sin\theta)] \\ p &= 2VI\left[\left(\frac{1}{2}(1 - \cos(2\omega t))\right)\cos\theta - \frac{1}{2}\sin(2\omega t)\sin\theta\right] \\ p &= VI\cos\theta - VI[(\cos(2\omega t)\cos\theta - \sin(2\omega t)\sin\theta)] \\ p &= VI\cos\theta - VI(\cos(2\omega t + \theta)) \end{aligned}$$

Hence, p consists of a real part $P = VI\cos\theta$ and a sinusoidal part with frequency twice as V and I .

4. Real Power: $P = VI\cos\theta$ (W)

Reactive Power: $Q = VI\sin\theta$ (VAR)

Apparent Power: $S = VI = \sqrt{P^2 + Q^2}$ (VA)

Complex Power: $\bar{S} = \bar{V}\bar{I}^* = P + jQ$

Power Factor: $PF = \cos\theta$

5. Question: If $P = 800W$, $pf = 0.8$ (lead), and $V = 200V$, find I , S , Q , z .

$$P = VI\cos\theta \implies 800 = (200)I(0.8) \implies I = 5A$$

$$Q = VI\sin\theta = (800)(6)(\sqrt{1 - 0.8^2}) = 600VAR$$

$$S = \sqrt{P^2 + Q^2} = \sqrt{800^2 + 600^2} = 1000VA$$

$$V = Iz \implies 800 = 5z \implies z = 40\Omega$$

2.4 3- ϕ Systems:

1. These are either 3 wire system (a, b, c) or 4 wire system (a, b, c, n)
2. Size of 1 - ϕ System > Size of 3 - ϕ System (for same rating)
3. **Phase Voltages in Star Systems:** V_{an}, V_{bn}, V_{cn}
4. **Phase Voltages in Delta Systems:** V_{ab}, V_{bc}, V_{ca}
5. **Line Voltages:** V_{ab}, V_{bc}, V_{ca}
6. In case of voltages, phase voltages (delta) = line voltages
7. **Phase Currents in Star Systems:** I_a, I_b, I_c
8. **Phase Currents in Delta Systems:** I_{ab}, I_{bc}, I_{ca}
9. **Line Currents:** I_a, I_b, I_c
10. In case of voltages, phase currents (star) = line currents
11. In 3- ϕ systems, instantaneous power is constant at all times.
12. **Balanced Set:** All voltages are 120° phase-shifted, and are in same amplitude and frequency.
13. **For Star Loads:** $V_L = \sqrt{3}V_{ph}$ and $I_L = I_{ph}$
14. **For Delta Loads:** $V_L = V_{ph}$ and $I_L = \sqrt{3}I_{ph}$
- 15.

$$P = \sqrt{3}V_L I_L \cos\theta$$

$$Q = \sqrt{3}V_L I_L \sin\theta$$

$$S = \sqrt{3}V_L I_L$$

3 Magnetic Circuits:

Consider a conductor with a ferromagnetic core, wrapped in coils of N turns carrying i current, then: $\oint H dl = Ni$.
 Ni is called the "magneto-motive force"

$$\vec{B} = \mu \vec{H} \implies \phi = \frac{1}{R} (mmf) = \frac{\mu A}{l} (mmf)$$

Now,

$$\lambda = N\phi \implies L = \frac{\lambda}{i} = \frac{N\phi}{i} = \frac{N^2}{R}$$

Question: Consider a magnetic circuit with $i = 1A$, $N = 400$ turns, $\mu_r = 3000$, $l_c = 50cm$, $l_g = 1mm$, $\mu_0 = 4\pi \times 10^{-7}$, and $A = 15cm^2$. Find L.

Solution:

$$R_c = \frac{l_c}{\mu A} = \frac{50 \times 10^{-2}}{(4\pi \times 10^{-7})(15 \times 10^{-4})} = 88.42 kAt/W_b$$

$$R_g = \frac{l_g}{\mu A} = \frac{1 \times 10^{-3}}{(3000)(4\pi \times 10^{-7})(15 \times 10^{-4})} = 520.9 kAt/W_b$$

$$L = \frac{N^2}{R_c + R_g} = \frac{(400)^2}{(88.42 + 520.9) \times 10^3} = 0.262 mH$$

3.1 Transformers and DC Machines:

Refer to the PPTX/PDF in the ../References directory.